

Data Analysis Project

(Due: Sunday, August 2, 2026, upload to Canvas by midnight)

This is a group project. That is, you will work on the project in groups, with 3 to 5 students in each group – you choose the group members by yourselves (you may consider choosing group members with different strengths, such as one being good in writing, one being good in R, one being good in data analysis, etc). **Each group should work *independently*, and should have a leader.** Each group decides the group leader by group members. Each group submits one data analysis report – see details below.

On the cover page of your report, please write the name and student number of each group member, the group leader, and indicate clearly the roles and contributions each group member makes in the project. *If you use AI in any part of the project, you must clearly indicate it in your report.*

Each group will need to find a dataset by yourselves (e.g., from internet or books). You should analyze the (same) dataset using at least **two different models or methods** learned in this course, and then compare and report your results and conclusions. For example, suppose that you wish to investigate if the times spent on study may partially explains final exam scores, adjusted for gender and major. You could fit a multiple linear regression model on the final exam scores. You could also check if study time may be related to whether the exam scores being over 90% or not, using a logistic regression model. Such an approach would give us additional insights. In general, data analysis should be conducted based on different models/methods to see if the conclusions are consistent or not.

Your report should contain the following materials:

- Introduction: this section should describe the background and objectives of the study,
- Data description: this section should describe where the dataset comes from, the sample size, how the data were collected, etc.
- Exploratory data analysis (EDA): this section may include summary statistics (e.g., means and standard deviations of the variables), graphical displays of the data, and any preliminary conclusions. No models are assumed in this section.
- Models: this section should include at least two statistical models for the same dataset, *justifications* of the choices of the models, model diagnostics, parameter estimates and their standard errors and p-values, and *interpretation* of the results.
- Conclusions and discussion: this section should provide overall conclusions, any limitations in the datasets and analyses, and validity of the conclusions.
- References
- Appendix.

Requirements:

- Your report is limited to **6 to 10 pages** (12 point font size), including tables and figures but excluding Appendix and cover page. Your report should be typed (no hand writing).
- You should summarize and interpret your computer outputs. Do *not* put raw R computer outputs and results in the main text of your report. Please put R code, additional tables/figures in the Appendix (it does not count for page limit).
- Each group should work **independently**, and list the **contributions of each group member**.

Submissions:

Each group should submit just **one report** (in pdf file) to Canvas by the **group leader**.

File formats:

Please use the following file format: **Stat301-project-Lang-Wu**, where Lang Wu is the name of the group leader.